Ch # 6 : Counting.

1) Product Rule
2) Addition Rule
3) Subtraction Rule
4) Division Rule

1) Product Rule :-

If a task is done in multiple ~~choice~~ steps and each step has a number of choices, the total number of ~~choices~~ ways is the product of choices for each step.

General Rule: If step 1 can be done in m ways and step 2 in n ways, total

$$= m \times n$$

Ex 1: Outfit Combinations

- 4 shirts
- 3 pants

Q How many outfits can you make?

$4 \times 3 = \boxed{12}$ outfits

Ex 2: 3-letter passwords

Each character can be any of 26 letters (A-Z).
Characters can repeat

$$26 \times 26 \times 26 = \boxed{17,576} = 26^3$$

Characters can't repeat.

$$26 \times 25 \times 24 =$$

Example 3: License Plates

Format: 2 letters followed by 3 digits.

$$26 \times 26 \times 10 \times 10 \times 10 = 676,000$$

Example 4: Meal Selection

You choose:

- 1 starter (4 options)
- 1 main course (6 options)
- 1 dessert (3 options)

$$4 \times 6 \times 3 = 72$$

2). Addition Rule :-

If a task can be done either in one of n_1 ways or in one of n_2, where none of the set of n_1 ways is the same as the n_2 ways, then there are $n_1 + n_2$ ways to do the tasks.

General Rule: If A can be done in m ways and B in n ways, and they are mutually exclusive, then:

$$= m + n$$

Example 1 : Choose Transport

- 3 ways to go by bus.
- 2 ways by train.

$$3 + 2 = \boxed{5 \text{ options}}$$

2 : Number of 1-digit or 2-digit numbers.

1-digit : 1-9 = 9
2-digit : 10-99 = 90

$$9 + 90 = \boxed{99}$$

3. Choose drink or Dessert :-

5 drinks
4 desserts
(not both)

$$5 + 4 = \boxed{9}$$

3) Subtraction Rule :-

If a task can be done either in one of n_1 ways or in one of n_2 ways, then the total numbers of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways. Also known as, the principle of inclusion - exclusion.

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Examples :-

1) In a class:

- 15 likes math
- 10 like english
- 4 like both

How many students like math or english?

$$15 + 10 - 4 = 21$$

2) In a school:

- 40 students play football
- 30 play cricket
- 12 play both

How many play football or cricket?

$$40 + 30 - 12 = 58$$

3) From numbers between 1 and 100, how many are divisible by 3 or 5 ?

$\frac{100}{3} = 33$, $\frac{100}{5} = 20$, $\frac{100}{15} = 6$

$33 + 20 - 6 = 47$

5) In class of 100 students :

70 passed Math
65 passed Science
40 passed both

How many passed atleast one subject ?

$70 + 65 - 40 = 95$

How many failed both ?

$100 - 95 = 5$

6) In a group a) 100 like pizza b) 60 like pasta
c) 30 like both d) 10 like neither.

Q) How many total are in the group ?

$100 + 60 - 30 + 10 = \boxed{140}$

4) Division Rule :-

If a task can be done in n ways, but each outcome is counted exactly k times due to repetition, symmetry, or ordering then the actual number of distinct outcome is.

$$\text{Distinct Outcomes} = \frac{n}{k}$$

Why do we divide ?

Because sometimes we overcome the number of outcomes by considering repeated or symmetric versions as different, even when they are actually the same.

Examples 1 : Rearranging Letters :-

Q.1 How many distinct ways can you rearrange the letter of the word "MOM"?

- Total letter = 3
- All permutations = $3! = 6$
- But letter M appear twice.

$$\frac{3!}{2!} = \frac{6}{2} = \boxed{3}$$

2) Circular Seating :-

Q: In how many ways can 4 friends sit around a round table?

. Total linear arrangement = $4! = 24$
. But in circle, all rotations are considered the same.
final answer:

$$\frac{n!}{n} = (n-1)! \Rightarrow (4-1)! = 3! = \boxed{6}$$

3) Repeated ~~Strings~~ Digits in Strings :-

Q How many 3-digit numbers can be formed using the digits 1, 1, 2?

. Total permutations $= 3! = 6$.
. But 1 appears twice, so divide by $2!$

$$\frac{3!}{2!} = \frac{6}{2} = \boxed{3}$$ 112, 121, 211

Q How many distinct permutations of the word "BALLOON".

BALLOON total letters = 7
Repeating = L×2, O×2

$$\frac{7!}{2!\,2!} = \frac{5040}{4} = 1260$$

Combining the Sum & Product Rule :-

Q1 Choosing an outfit

A person has 2 jackets and 3 shirts. If they decide to wear only a shirt, or a combination of one jacket and one shirt, how many choices do they have?

Solution

- Case 1 (only shirts) : 3 shirts
- Case 2 (Jacket + Shirt): $2 \times 3 = 6$ Combination

- Total Choices : $3 + 6 = \boxed{9}$

Q2 Password Formation

A password must be either :

A 2-digit number, OR
A 1-letter + 1-digit code (letter from A to Z)

How many such passwords are possible?

Solution

Case 1 (2 digit number):

$10 \times 10 = 100$

Case 2 (1 digit + 1 letter):

$10 \times 26 = 260$

Total Pass $= 260 + 100 = \boxed{360}$

Permutations

- A permutation is an arrangement of items where order matters.

$$P(n, r) = \frac{n!}{(n-r)!}$$

- n = total items
- r = items you want to arrange.

$${}^nP_1 = n \; ; \quad {}^nP_n = n! \; ; \quad {}^nP_{n-1} = n!$$

Q: How many ways can you arrange 3 students from a group of 10 for 1st, 2nd, 3rd positions?

sol:

$$P(10, 3) = \frac{10!}{(10-3)!} = \frac{10 \times 9 \times 8 \times 7!}{7!} = 720 \text{ Ways}$$

2. Permutations where all items are arranged ($nP_n = n!$) :-

Q How many ways to arrange 5 books on a shelf?

sol:

$$P(5, 5) = 5! = \boxed{120}$$

Q5 Statement Labels: Suppose statement labels in a programming language can be either a single letter or a letter followed by a digit. Find the number of possible labels.

Solution

Use the product Rule.

$$26 + (26 \cdot 10) = \boxed{286}$$

Q6 Counting Passwords :-

Q Each user on a computer system has a password, which is 6 to 8 characters long where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

$$P = P_6 + P_7 + P_8 :$$

$$P_6 = 36^6 - 26^6$$
$$P_7 = 36^7 - 26^7$$
$$P_8 = 36^8 - 26^8$$

Q3 Food Combo Offer

A restaurant offers:

- 3 types of burgers
- 4 types of pizzas
- You can choose only a burger, only a pizza or one of each.

Q How many different meal options are available?

Sol:

- Only burgers = 3
- Only Pizza = 4
- One burger + one pizza = 3×4 = 12
- Total Options = 3+4+12 = **19**

Q4 Selecting Course

A student can choose:

- One course from CS (5 options).
- One course from Math (4 options).
- OR one course from each department.

How many total ways can the student make a choice?

Sol: Only CS = 5

Only Math = 4

One from each = 5×4 = 20

Total = 5+4+20 = **29**

3. Permutations With Repetition :-

You are allowed to reuse items (order matters, repetition allowed).

$$\text{Total Ways} = n^r$$

where,

n = number of available choices

r = number of positions to fill

Q How many 4-digits numbers can be formed using digits 0-9 (repetition allowed)?

$$10^4 = 10,000$$

Q License plates with 3 letters followed by 3 digits (repetition allowed)?

$$26^3 \times 10^3 = 17,576,000 \text{ plates}$$

4. Permutations with Repeated Letters.

When some items are indistinguishable (repeated).

$$\text{Total Ways} = \frac{n!}{r_1! \times r_2! \times \ldots}$$

Q How many distinct arrangements of word BALLOON?

- Letters : 7 total
- Repeated : L(2), O(2)

$$= \frac{7!}{2! \times 2!} = \frac{5040}{4} = \boxed{1260}$$

Q How many unique arrangements of the word MISSISSIPPI?

- letters : 11 total
- Repeats : I(4), S(4), P(2)

$$= \frac{11!}{4!\,4!\,2!} = \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4!}{4 \times 3 \times 2 \times 1 \times 4! \times 2 \times 1}$$

$$= 11 \times 10 \times 9 \times 7 \times 5$$

$$= 110 \times 9 \times 35$$

$$= 990 \times 35$$

$$= \boxed{34650}$$

5. Circular Permutations :-

Used when objects are arranged in a circle instead of a line.

- No of rotation or direction matters :

$$(n-1)!$$

- If necklace/garland (also mirror images same) :

$$\frac{(n-1)!}{2}$$

Q How many ways to arrange 6 friends around a round table?

$$= (6-1)! = 5! = \boxed{120}$$

Q: How many ways to make a ~~chain~~ necklace of 6 beads (rotations and reflections same)?

$$= \frac{(6-1)!}{2} = \frac{120}{2} = \boxed{60}$$

Q From 6 men & 4 women, choose 3 people including at least 1 woman.

sol:

Cases:

- 1 women, 2 men : $C(4,1) \times C(6,2) \rightarrow a$
- 2 women, 1 men : $C(4,2) \times C(6,1) \rightarrow b$
- 3 women : $C(4,3) \rightarrow c$

Add all cases : $a + b + c$

4) Dividing into Groups :-

Distribute people into teams.

Q Divide 6 people into 2 teams of 3 people each.

sol:

$$\frac{C(6,3)}{2!} = \frac{20}{2} = \boxed{10 \text{ Ways}}$$

- Divide by 2! to remove identical team ordering.

5) Selecting Items with Restrictions :-

Q From 10 students, choose 4 such that Ali is always included.

Fix Ali → choose 3 from remaining 9.

$$C(9,3)$$

1) Selecting r items from n items :-

Q: From 8 books, choose 3 to gift.

$$C(8,3) = \frac{8!}{3!\,5!} = \frac{8\times7\times6\times5!}{3!\,5!} = \boxed{56 \text{ ways}}$$

2) Committee Formation Problems :-

* Choosing people for team, committee, board, group, etc.

Formula:

$C(n,r)$ (selecting r people from n).

Q: From 12 people, choose 5 to form a committee.

$$C(12,5) = \frac{12!}{5!\,7!} = \boxed{792 \text{ ways}}$$

3) Selecting from groups (men, women, etc) With Conditions :-

Seperate groups (e.g., at least 2 women).

Formula: (multiply combinations per group).

Combinations

When order does not matter, we use Combinations instead of permutations.

Example:-

Choosing 3 friends to go on a trip from a group of 6.
Doesn't matter if its (Ali, Areeb, Aman) or (Aman, Areeb, Ali) - same group.

$$C(n,r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

- n = total items
- r = items to choose
- Order doesn't matter

Properties	Meaning
$C(n,r) = C(n,n-r)$	Symmetry
$C(n,0) = 1$	Choosing Nothing
$C(n,1) = n$	Choosing one
$C(n,n) = 1$	Choosing all
$C(n,n-1) = n$	Choosing all but one

Ex #6.3

10) There are six different candidates for governor of a state. In how many orders can the names of the candidates can be printed on a ballot?

solve

$$P(6,6) = 6! = 720$$

11) How many bit strings of length 10 contain

a) exactly four 1s?
b) at most four 1s?
c) at least four 1s?
d) an equal number of 0s and 1s?

a) solve

A bit string of length 10 contains exactly 4 ones and 6 zeroes.

We need to choose 4 positions out of 10 to place 1s. The rest will automatically 0s!

$$C(10,4) = \frac{10!}{4!(10-4)!} = \boxed{210}$$

Ex #6.3

123, (132, 213, 231, 312, 321)
124, (142, 214, 241, 412, 421)
125, (152, 215, 251, 512, 521)
134, (143, 314, 341, 413, 431)
135, (153, 315, 351, 513, 531)
145, (154, 415, 451, 514, 541)
234, (243, 324, 342, 423, 432)
235, (253, 325, 352, 523, 532)
245, (254, 425, 452, 424, 542)
345, (354, 435, 453, 534, 543)

7) Find the number of 5-permutations of a set with nine-elements:

$$P(9,5) = 15,120$$

8) In how many different orders can five runners finish a race if no ties are allowed?

solve

$$P(5,5) = 5! = 120$$

9) How many possibilities are there for the win, place, and show (first, second, third) positions in a horse race with 12 horses if all orders of finish are possible?

$$P(12,3) = 1320$$

Ex #6.3

1) List all the permutations of $\{a, b, c\}$.

solution

abc
acb
bac
bca
cab
cba

2) How many different permutations are there of the set $\{a, b, c, d, e, f, g\}$?

solution

$$P(7,7) = 7! = 5040$$

3) How many permutations of $\{a, b, c, d, e, f, g\}$ end with a?

$$\underline{\quad}\times\underline{\quad}\times\underline{\quad}\times\underline{\quad}\times\underline{\quad}\times\underline{\quad}\times\underline{a}$$

$$6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$P(6,6) = 6! = 720$$

4) Let $S = \{1, 2, 3, 4, 5\}$

a) List all the 3-permutations of S.
b) List all the 3-combinations of S.

Ex #6.3

d) solve

$C(12, 6) = \boxed{924}$

13) A group contain n men and n women. How many ways are there to arrange these people in a row if the men and women alternate?

Start with man Case 1:

no of ways to arrange men $= n!$
no of ways to arrange women $= n!$
so total no for this case $= n! \cdot n!$

Start with woman Case 2:

Same logic applies just switched starting positions.

$n! \cdot n!$

Add Case 1 and Case 2.

$2(n!)^2$

b) they don't alternate.

$(2n)!$

Ex #6.3

d) An equal number of 0s and 1s?

A bit string of length 10 has equal number of 1s and 0s only when there are 5 ones and 5 zeroes.

$$C(10,5) = \frac{10!}{5! \cdot 5!} = \boxed{252}$$

12) How many bit strings of length 12 contain

a) exactly 3 1s?
b) at most 3 1s?
c) at least 3 1s?
d) an equal number of 0s and 1s?

a) solve

$$C(12,3) = \boxed{220}$$

b) solve

$$= C(12,0) + C(12,1) + C(12,2) + C(12,3)$$
$$= 1 + 12 + 66 + 220$$
$$= \boxed{299}$$

c) solve

$$= 2^{12} - (C(12,0) + C(12,1) + C(12,2))$$
$$= 4096 - (1 + 12 + 66)$$
$$= 4096 - 79 = \boxed{4017}$$

Ex # 6.3

b) At most four 1s ?

means the number of 1s can be : 0, 1, 2, 3, 4.
We add all possibilities from 0 to 4.

$$\text{Total} = \sum_{K=0}^{4} P(10, k)$$

$$= P(10,0) + P(10,1) + P(10,2) + P(10,3) + P(10,4)$$

$$= 1 + 10 + 45 + 120 + 210$$

$$= \boxed{386}$$

c) At least four 1s ?

Means 4 or more : 4, 5, 6, 7, 8, 9, 10 ones.

$$\text{Total} = \sum_{k=4}^{10} P(10, k)$$

$$= P(10,4) + P(10,5) + P(10,6) + P(10,7) + P(10,8) + P(10,9) + P(10,10)$$

$$= 210 + 252 + 210 + 120 + 45 + 10 + 1$$

$$= \boxed{848}$$

Ex # 6.3

14) In how many ways can a set of two positive integers less than 100 be chosen?

sol:
We are simply choosing 2 distinct integers out of 99.

$$C(99,2) = \frac{99!}{2!\,97!} = \frac{99 \times 98 \times 97!}{2 \times 97!}$$

$$= 99 \times 49$$

$$= 4851$$

15) In how many ways can a set of five letters be selected from the english alphabet?

sol:
We use Combination because talking about sets which does not follow any order.

$$C(26,5) = 65780$$

16) How many subsets with an odd no of elements does a set with 10 elements have?

$$= C(10,1) + C(10,3) + C(10,5) + C(10,7) + C(10,9)$$
$$= 10 + 120 + 252 + 120 + 10 = 512$$

17)

Ex #6.3

Since we know total no of subsets $= 2^n$.

$2^{10} = 1024$

and there are same, equal no of subsets of odd no of elements and even no of elements, so

$1024/2 = \boxed{512}$

17) ~~How many~~ How many subsets with more than two elements does a set with 100 elements have?

Sol:

A set with n elements has a total of 2^n subsets, including the empty and full set.

2^{100} = total subsets.

We want to exclude all subsets that have 0, 1 or 2 elements.

so, we subtract

$C(100, 0) = 1$

$C(100, 1) = 100$

$C(100, 2) = 4950$

total $= 5051$

$\boxed{= 2^{100} - 5051}$

Ex # 6.3

18) A coin is flipped eight times where each flip comes up either heads or tails. How many possible outcomes.

a) are there in total ?
b) contain exactly 3 heads ?
c) contain ~~exactly~~ ~~3~~ ~~to~~ atleast 3 heads ?
d) contain the same of heads and tails ?

a) sol:

Total outcomes $= 2^8 = \boxed{256}$

b) sol:

We are choosing 3 flips out of 8 to be heads.
This is a combination problem.

$$C(8,3) = \boxed{56}$$

c) sol:

"At least 3 heads" means:

3 heads
4 heads
5 heads
6 heads
7 heads
8 heads

so we add:

Ex #6.3

f) the strings CBA and BED?

B is in both and can't be appear in any permutation. so

Answer = 0

22) How many permutations of the letters ABCDEFGH contain

a) the string ED ?

[ED], A, B, C, F, G, H → 7 blocks

7! = 5040

b) contain the string CDE ?

[CDE], A, B, F, G, H → 6 blocks

6! = 720

c) the strings BA and FGH ?

[BA], [FGH], C, D, E → 5 blocks

5! = 120

Ex # 6.3

21) How many permutations of letter ABCDEFG contain

a) the string BCD ?

Treat "BCD" as one block, so:

$$\underbrace{[BCD]}_{1}, \underbrace{A}_{2}, \underbrace{E}_{3}, \underbrace{F}_{4}, \underbrace{G}_{5}$$

$5! = \boxed{120}$

b) the string CFGA ?

$$\underbrace{[CFGA]}_{1}, \underbrace{B}_{2}, \underbrace{D}_{3}, \underbrace{E}_{4} \qquad 4! = \boxed{24}$$

c) the strings BA and GF ?

$[BA], [GF], C, D, E \qquad 5! = \boxed{120}$

d) the strings ABC and DE ?

$[ABC], [DE], F, G \qquad 4! = \boxed{24}$

e) the string ABC and CDE ?
both C is overlapping so it only exist when ABCDE is together.

$[ABCDE], F, G \qquad 3! = \boxed{6}$

Ex # 6.3

d) contain the same number of heads and tails?

$C(10,5) = \boxed{252}$

20) How many bit strings of length 10 have

a) exactly 3 0s?

$C(10,3) = \boxed{120}$

b) more 0s than 1s?

Since total bits = 10, this happens when the number of 0s is 6, 7, 8, 9, 10

Lets add:

$= C(10,6) + C(10,7) + C(10,8) + C(10,9) + C(10,10)$
$= 210 + 120 + 45 + 10 + 1$
$\boxed{= 386}$

c) at least seven 1s?

$= C(10,7) + C(10,8) + C(10,9) + C(10,10)$
$= 120 + 45 + 10 + 1 = \boxed{176}$

d) at least 3 1s?

solve

$$\sum_{k=3}^{10} C(10,k) = C(10,3) + \dots + C(10,10)$$
$$= \boxed{968}$$

24) How many ways are there for 10 women and 6 men to stand in a line so that no two men stand next to each other? [hint: First position the women and then consider possible positions for the men.]

Solution

_W_W_W_W_W_W_W_W_W_W_

Women arrangement ways = 10!

6 men can be placed in 11 gaps in

C(11, 6) ways.

And can be arrange among themselves in

6! ways.

So:

Total Ways = 10! × C(11, 6) × 6!

= 1,207,084,032,000

We can also just

P(10, 10) × P(11, 6)

We first place the 8 men in line.
This can be done in:

$8!$ ways

Once the 8 men are placed, the create 9 possible gaps where women can be placed:

_ M _ M _ M _ M _ M _ M _ M _ M _

To ensure no two women are adjacent, we must place one women per gap, and we must select 5 out of these 9 gaps.

This can be done in:

$C(9,5)$ ways

The 5 women can be arranged among themselves in:

$5!$ ways

Final Answer:

Total Ways $= 8! \times C(9,5) \times 5!$

$= 60\,963\,8400$

Ex# 6.3

d) the strings AB, DE and GH?

[AB], [DE], [GH], C, F → 5 blocks

$5! = 120$

e) the strings CAB and BED?

[CABED], F, G, H → 4 blocks

$4! = \boxed{24}$

f) the strings BCA and ABF?

[BCA] [ABF] cant possible in one string.

$\boxed{\text{Answer} = 0}$

23) How many ways are there for eight men and five women to stand in a line so that no two women stand next to each other?

Given:

Total = 8 men (M) and 5 women (W)

Condition = No. to women should be adjacent.

27) One hundred tickets, numbered 1, 2, 3, ..., 100, are sold to 100 different people for a drawing. Four different prizes are awarded, including a grand prize (a trip to Tahiti). How many ways are there to award the prizes if

a) there are no restrictions?

We are choosing 4 distinct winners in order from 100 people

Total Ways = $P(100, 4) = 94109400$

b) the person holding ticket 47 wins grand prize?

We fix ticket 47 for the grand prize.
Now choose and arrang the remaining 3 prizes among the remaining 99 people.

Total Ways = $P(99, 3) = 99 \times 98 \times 97$
$= 941,094$

c) the person holding ticket 47 wins one of the prize?

47 can win 4 diff prizes → 4 choices.
Choose and arrange the remaining 3 prizes.

Total Ways = $4 \times P(99, 3)$
$= 941,094 \times 4 = 3,764,376$

Arrange women within the block.

5!

Total Ways = 5! × 5!

=

26) How many ways are there for three penguins and six puffins to stand in a line so that:

a) All puffins stand together

b) All penguins stand together.

25) How many ways are there for 4 men and 5 women to stand in a line so that

a) all men stand together
b) all women stand together.

a) Treat the 4 men as a single unit (block), so now we are arranging.

- 1 block of men
- 5 individual women

So we have 6 units to arrange:

$6!$

Men are distinct. They can be arranged among themselves in:

$4!$

Total ways = $6! \times 4! =$

b) Now treat all 5 women as a block.

- 1 block women
- 4 men

$5!$

d) The person holding ticket 47 does not win a prize?

We choose and assign all 4 prizes from the remaining 99 people.

$= P(99, 4) = 99 \times 98 \times 97 \times 96$

$= 90,345,024$

e) the people holding 19 and 47 both win prizes?

first 19 one can win 4
second 47 one can win 3
the other 98 can win 2

so, $= 4 \times 3 \times P(98, 2)$

$= 114,072$

f) the people holding 19, 47, and 73 all win prizes?

$= 4 \times 3 \times 2 \times 97$

$= 2328$

g) the people holding tickets 19, 47, 73, 97 all win prizes?

$= 4 \times 3 \times 2 \times 1$

$= 24$

h) none of the people holding 19, 47, 73, 97 wins a prize?

$= P(96, 4) = 96 \times 95 \times 94 \times 93$

$=$

i) the grand prize winner is a person holding ticket 19, 47, 73, 97?

: Grand prize winner $\Rightarrow$ 4 ways.
Step 2 same:

$= 4 \times P(99, 3) = 4 \times 99 \times 98 \times 97$
$= 3,764,376$

j) the people holding 19, 47 win prizes, but the people holding 73, 97 do not win prize?

$= 4 \times 3 \times P(96, 2)$
$= 12 \times 96 \times 95$
$= 109,440$

28) Thirteen people on a softball team show up for a game.

a) How many ways are there to choose 10 players to take the field?

$= C(13, 10) = 286$

b) How many ways are to assign 10 positions by selecting players from the 13 who show up?

We need to worry about the order now.

$= P(13, 10) = 13 \times 12 \times 11 \times 10 \ldots \times 3$

$= 1,037,836,800$

c) Of the 13 people show up 3 are women. How many ways are there to choose 10 players to take the field if atleast one of these players must be a women?

total ways $= C(13, 10)$

no women case $= C(10, 10)$

Final $= C(13, 10) - C(10, 10)$

$= 286 - 1$

$= 285$

39) A Club of 25 members.

a) How many ways are there to choose 4 members of the club to serve on an executive committee?

$= C(25, 4)$

$= 12,650$

Ex # 6.1

1) a) $18 \times 325 = 5850$
b) $18 + 325 = 343$

2) $= 27 \times 37$
$= 999$

3) a) $4 \times 4 \times 4 \times \cdots \times 4 = 4^{10} = 1,048,576$
b) $5 \times 5 \times 5 \times \cdots 5 = 5^{10} = 9,765,625$

4) Product Rule: $12 \times 2 \times 3 = 72$ shirt types

5) Product Rule $6 \times 7 = 42$ diff pairs

6) Product Rule $6 \times 4 = 24$ routes

7) Product Rule $26^3 = 17,576$

8) Product Rule $26 \times 25 \times 24 = 15,600$

9) Product Rule $1 \times 26 \times 26 = 676$

10) Each bit can 0 or 1 · so $2^8 = 256$

11) $2^8 = 256$

12)

13) $n + 1$ (counting the empty string)

14) 2^{n-2}

a) Total committees without restrictions = $C(16, 5)$
Committees with no women = $C(9, 5)$

Committees with at least one woman =

$= C(16, 5) - C(9, 5)$
$= 4368 - 126 = 4242$

b) This time we want to exclude both extreme cases.

- All 5 are women = $C(7, 5)$
- All 5 are men = $C(9, 5)$

Total Committees = $C(16, 5)$

$= C(16, 5) - C(7, 5) - C(9, 5)$
$= 4368 - 21 - 126$
$= 4221$

d) How many ways are there to choose a president, vice president, secretary, and treasurer of the club, where no person can hold more than one office?

$$P(25, 4) = 25 \times 24 \times 23 \times 22$$
$$= 303,600$$

30)

The order doesn't matter in arranging letter T/F

$$= C(40, 17)$$
$$= 88,732,378,800$$

32)

28) $\underline{10}\ \underline{10}\ \underline{10}\ \underline{26}\ \underline{26}\ \underline{26} + \underline{26}\ \underline{26}\ \underline{26}\ \underline{10}\ \underline{10}\ \underline{10}$

$= (10^3 \times 26^3) + (10^3 \times 26^3)$

$= 2(10^3 \times 26^3)$ Ans

29) $\underline{26}\ \underline{26}\ \underline{10}\ \underline{10}\ \underline{10}\ \underline{10} + \underline{10}\ \underline{10}\ \underline{26}\ \underline{26}\ \underline{26}\ \underline{26}$

$(26^2 \times 10^4) + (10^2 \times 26^4)$

30) $\underline{26}\ \underline{26}\ \underline{26}\ \underline{10}\ \underline{10}\ \underline{10} + \underline{26}\ \underline{26}\ \underline{26}\ \underline{26}\ \underline{10}\ \underline{10}$

$(26^3 . 10^3) + (26^4 . 10^2)$

31) $\underline{26}\ \underline{26}\ \underline{10}\ \underline{10} + \underline{26}\ \underline{26}\ \underline{10}\ \underline{10}\ \underline{10} +$
$\underline{26}\ \underline{26}\ \underline{26}\ \underline{10}\ \underline{10} + \underline{26}\ \underline{26}\ \underline{26}\ \underline{10}\ \underline{10}\ \underline{10} +$

$= (26^2 . 10^2) + (26^2 . 10^3) + (26^3 . 10^2) + (26^3 . 10^3)$

$=$

25) a) first find total strings $= 10^3 = 100$

Now subtract the string where all three digits are same. 111, 222, 333, ..., 000

Ans $= 1000 - 10 = \boxed{990}$

d) $= \underline{5}\ \underline{10}\ \underline{10} = 5 \times 10^2 = 500$

c) Choose 2 position out of 3.

$C(3,2) = 3$

44x, 4x4, x44

Choose digit for 3rd position → 9 choices

$3 \times 9 = \boxed{27}$

26) a) $\underline{10}\ \underline{9}\ \underline{8}\ \underline{7} = \boxed{5040}$

b) $10^3 \times 5 = \boxed{5000}$

c) $C(4,3) = 4$ then $4 \times 9 = \boxed{36}$

27) 3^{50}

DNA bases: A, T, C, G → 4 choices

RNA bases: A, U, C, G → 4 choices

15) length : 1, 2, 3, 4

Number of strings for length $k = 26^k$

Total:

$26^1 + 26^2 + 26^3 + 26^4 = 475,254.$

16) Total number of strings $= 26^4$

Number of 4 letter strings without $x = 25^4$

So, string with at least 1 $x = 26^4 - 25^4$

$= 456,976 - 390,625$

$= 66,351$

17) $128^5 - 127^5 =$

18) a) $4^4 = 256$

b) $4^3 = 64$

c) $2^5 = 32$

d) $3^5 = 243$

19) a) $3^6 = 729$

b) $4^4 = 256$

c) $4^5 = 1024$

d) $2^6 = 64$

26)

20, 21, 22, 23, 24 are of Divisible.

27)